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Longitudinal hippocampal axis in large-scale cortical systems underlying development and episodic memory

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The hippocampus is functionally specialized along its longitudinal axis with intricate interactions with cortical systems, which is crucial for understanding development and cognition. Using a well-established connectopic mapping technique on two large resting-state functional MRI datasets, we systematically quantified topographic organization of the hippocampal functional connectivity (hippocampal gradient) and its cortical interaction in developing brains. We revealed hippocampal functional hierarchy within the large-scale cortical brain systems, with the anterior hippocampus preferentially connected to an anterior temporal (AT) pathway and the posterior hippocampus embedded in a posterior medial (PM) pathway. We examined the developmental effects of the primary gradient and its whole-brain functional interaction. We observed increased functional specialization along the hippocampal long axis and found a general whole-brain connectivity shift from the posterior to the anterior hippocampus during development. Using phenotypic predictive modeling, we further delineated how the hippocampus is differentially integrated into the whole-brain cortical hierarchy underlying episodic memory and identified several key nodes within PM/AT systems. Our results highlight the importance of hippocampal gradient and its cortical interaction in development and for supporting episodic memory.

hippocampus | gradient | functional connectivity | episodic memory | development

Significance

Despite evidence for the human hippocampus being functionally specialized along a longitudinal gradient, how the longitudinal gradient interacts with the rest of the brain in developing brains and its implication for development and cognition remain unclear. Leveraging large-scale resting-state functional MRI datasets, we identified the primary gradient of connectivity variation within the hippocampus and revealed how the longitudinal hippocampal gradient is embedded within the whole-brain cortical system. We further demonstrated the critical role that hippocampal gradient and its cortical interaction play in development and episodic memory.

The hippocampus is a complex structure in the medial temporal lobe implicated in various neuropsychological functions, especially memory (1). One of the most researched questions about the hippocampus concerns the functional specialization along its longitudinal axis from the head (anterior) to its tail (posterior). Earlier activation studies using positron emission tomography and functional MRI (fMRI) have shown a dissociation between anterior and posterior hippocampus for episodic memory encoding and retrieval, respectively (2, 3). However, neither anterior nor posterior hippocampus functions independently but interacts with various key brain networks to perform cognitive processes such as memory (4). A recent study using densely sampled adult fMRI data suggested dual hippocampal-parietal circuits that subserved self- vs. goal-oriented processing (5). They showed the anterior hippocampus was functionally coupled with the default mode network (DMN), while the posterior hippocampus was more connected with the posterior medial (PM) parietal cortex, which is adjacent to the DMN and previously considered part of the DMN. The anterior and posterior hippocampus are also hypothesized to be differentially embedded within an anterior temporal (AT) and PM cortical system (6). More posteriorly, the hippocampus interacts with medial and posterior parietal regions via the parahippocampal cortex, all of which are part of the PM system. Meanwhile, the anterior portion of the hippocampus interacts with areas in the lateral AT lobe and lateral orbitofrontal cortex via the perirhinal and retrosplenial cortex, forming the AT system. These two systems interact via the hippocampus to support the multidimensional episodic memory (7). A recent study revealed that hippocampal-cortical connectivity is dynamically reconfigured during episodic encoding retrieval, to facilitate a flexible dissection and recombination of gist-like and precision-related information (8). Moreover, Reagh and Ranganath showed different cortico-hippocampal networks preferentially represent contextual and entity information during naturalistic events (9). These recent studies highlight the need to study hippocampal functional specialization beyond the medial temporal lobes by including the large-scale cortical systems.

Although the dichotomous view of the hippocampal longitudinal axis (anterior vs. posterior) used to dominate the field, an alternative view of a continuous functional transition, also known as "gradient" is receiving increasing attention (10). Recent methodological advances (11, 12) have paved the way for investigating such smooth transition

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without the need for a priori (sometimes arbitrarily) definition of the subfield. A recent resting-state fMRI study showed a smooth connectivity gradient following the anterior-to-posterior hippocampus axis in healthy young adults (13). The longitudinal hippocampal gradient has since been replicated in various studies using task fMRI (14), tractography (15), gene expression (16), and multimodal data (17).

Despite the recent discoveries, knowledge gaps remain. First, although the longitudinal gradient is well established in adults, it remains to be investigated in children and its implication for development is unclear. Second, while previous gradient studies have furthered our understanding of hippocampal local organization and functional specialization, the hippocampus alone does not support complex behaviors like episodic memory. Therefore, it is critical to look beyond hippocampus and also study its whole-brain coupling pattern for a more holistic understanding. Third, it is yet to be determined how such hippocampal-cortical coupling patterns support individual differences in episodic memory. Addressing these questions may be an important first step to understand the differential integration of the hippocampus underlying development and key cognitive processes such as memory (18), executive function (19), and other high-order functions (20).

To fill these gaps, we utilized two large-scale publicly available resting-state fMRI datasets with typically developing children and adolescents between ages 8 and 21 (21, 22) to examine the longitudinal hippocampal gradient in developing brains. Using a well-established connectopic mapping analysis (11), we characterized the functional gradient that describes the gradual functional connectivity changes within the hippocampus and positioned the hippocampal gradient within the cortical hierarchy by measuring its whole-brain projection map and examined how they mature during development. We then sought to examine how longitudinal hippocampal gradient interacts with the rest of the brain to support key hippocampus-related cognitive functions such as episodic memory. To gain deeper insights into the neurobiological meaning of the hippocampal whole-brain coupling patterns, we combined gradient methodology with a multivariate predictive modeling framework to study how functional coupling between the primary hippocampal gradient and large-scale cortical systems supports individuals' episodic memory, and we further validated our model in an independent dataset.

Results

Our discovery dataset was Human Connectome Project in Development (HCP-D) (22), which consisted of 598 low-motion (mean framewise displacement <0.4 mm) typically developing children between ages of 8 and 21. The replication data were from the Nathan Kline Institute-Rockland Sample (NKI) (21) and this sample was composed of 196 low-motion typically developing individuals within the same age range (*SI Appendix*, Table S1 for demographic information). Details about participants, image acquisition and preprocessing, connectopic mapping, and predictive modeling can be found in the *Materials and Methods* section.

Longitudinal Hippocampal Gradients in Children Mimicking Hippocampal Subdivisions from Adults. Using the connectopic mapping technique (https://github.com/koenhaak/congrads), we characterized the gradients separately for the left and right hippocampus, given the evidence of their distinct functional roles (23). Briefly, we measured the whole-brain connectivity at each hippocampal voxel and generated a voxel-by-voxel within-hippocampus similarity matrix. A nonlinear dimensionality reduction technique—Laplacian Eigenmap—was then used to extract primary dimensions (i.e., gradients) that preserve the local connectivity structure within the hippocampus so that voxels with similar connectivity profiles were close to one another in the low-dimensional space (24). As shown in Fig. 1*A*, the primary hippocampal gradients followed the expected longitudinal axis and were highly similar across the discovery and replication datasets (r = 0.99). The second to fifth gradient can be found in *SI Appendix*, Figs. S1 and S2 (r = 0.96, 0.83, 0.79, 0.81, respectively). The longitudinal organization of the hippocampal gradient supports previous theoretical and empirical work on hippocampal organization (10, 13, 25). Moreover, we further validated the primary hippocampal gradient against the recently developed Melbourne subcortical atlas (26) (Fig. 1*B*). We first applied k-means clustering and set the number of clusters equal to four. We then compared the Dice index between the k-means clusters with four hippocampal subdivisions of the Melbourne subcortical atlas (Dice index = 0.69 ± 0.19). We observed that the primary gradient was well segregated along previously found hippocampal subdivisions (one-way ANOVA, $P < 0.001$), as shown in Fig. 1*C*. We also examined the anatomical gradients by performing principal component analysis on Montreal Neurological Institute (MNI) coordinates of the hippocampus, and the primary anatomical gradient was highly correlated with the hippocampal functional gradients (r = 0.98, $P < 0.001$). Taken together, our results confirmed that the longitudinal hippocampal gradient matched underlying functional and anatomical organization.

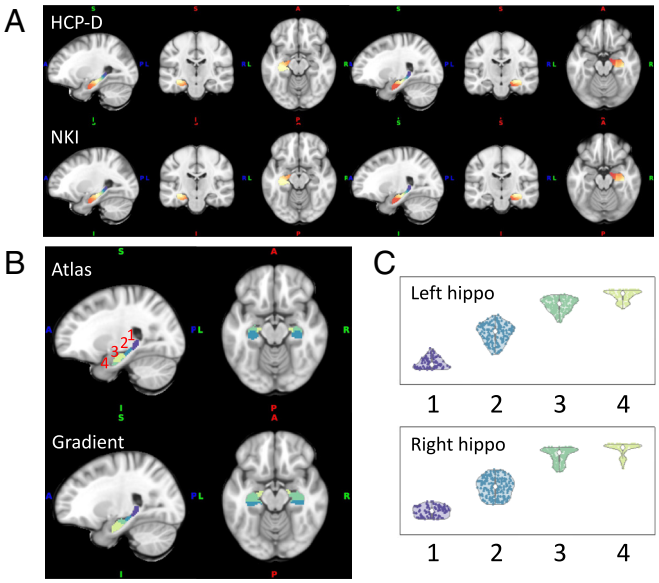

Fig. 1. (*A*) Primary hippocampal gradient for the left and right hippocampus in HCP-D and NKI datasets. (*B*) Comparing hippocampal gradient k-means clusters with subdivisions of Melbourne subcortical atlas, i.e., hippocampus head lateral and medial division, body, and tail. (*C*) Violin plots of the primary gradient distribution across four hippocampal subdivisions based on the Melbourne subcortical atlas, color-coded according to Melbourne subcortical atlas in panel *B*.

Hippocampal Projection Map Reflecting Changing Functional Connectivity Pattern along the Longitudinal Gradient. Next, we investigated the whole-brain projection maps for the primary longitudinal hippocampal gradient, which reflects the functional coupling patterns between the topologically varying gradient and the rest of the brain (27). Briefly, we first extracted the timeseries associated with the primary gradient, which was then regressed

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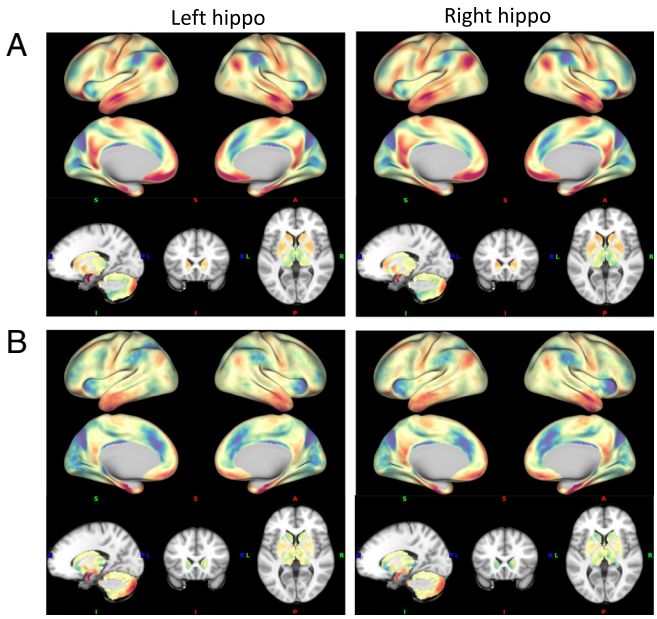

Fig. 2. Projection maps associated with the primary hippocampal gradient for the left and right hippocampus in HCP-D (*A*) and NKI datasets (*B*).

onto the rest of the brain using dual regression (28). As shown in Fig. 2, the resulting group-level projection maps corresponding to the primary gradients were similar across the two datasets ($r = 0.77$, $p_{spin} < 0.001$; 1,000 permutations) (29). The consistency of the projection maps for the second gradient was lower across datasets, with an overall similarity of 0.58 (*SI Appendix*, Fig. S3).

To further dissect the association between the projection map and longitudinal hippocampal gradients, we employed a moving-seed functional connectivity analysis. We calculated the seed-based functional connectivity of hippocampal subdivisions while moving the seed from the posterior to the anterior hippocampus (Fig. 3*A*). We then correlated the seed-based connectivity patterns with their relative location along the gradient to generate the r map (Fig. 3*B*), which described how connectivity changed along the longitudinal gradient. Regions with positive r values correspond to those with increasing connectivity as the seed moves from the posterior to the anterior hippocampus and vice versa. The r map was highly correlated with the primary projection maps with an averaged similarity of 0.92 measured by Pearson correlation. The r map confirmed that the projection map is a proxy of connectivity changes along the longitudinal hippocampal gradient. Hence, regions indicated by red circles in Fig. 3*A* were preferentially connected to the anterior hippocampus, including the angular gyrus, ventromedial prefrontal cortex, temporal pole, and amygdala. On the other hand, regions circled in blue were preferentially connected to the posterior hippocampus, including posterior parieto-occipital regions, inferior parietal lobule, lateral orbitofrontal cortex, anterior cingulate cortex, retrosplenial cortex, and precuneus.

We conducted a control analysis to ensure the projection maps were not driven by nuisance factors such as spatial autocorrelation (30). We found that the real data produced significantly more consistent across-subject projection maps than those from random data (real: $r = 0.20 \pm 0.09$; random $r = 0 \pm 0.05$; two-sample *t* test $P < 0.001$), as shown in *SI Appendix*, Fig. S4.

Age-Related Changes of the Longitudinal Gradient and Projection Maps. To examine how the functional organization along the hippocampus' longitudinal axis changes during development, we computed the Spearman correlation between the individuals' primary gradient with age, controlling for gender, head motion, and relative hippocampal volumes (normalized to the intracranial volume). Using the HCP-D dataset (n = 598), we found a positive aging effect on the anterior hippocampal gradient, while a negative effect on the posterior gradient (Fig. 4*A*). This age-related correlation map closely resembled the longitudinal gradients ($r_{Lhippo} = 0.69$, $r_{Rhippo} = 0.87$, Ps < 0.001), a finding replicated in the NKI dataset ($r_{Lhippo} = 0.68$, $r_{Rhippo} = 0.83$, Ps < 0.001, *SI Appendix*, Fig. S5).

Furthermore, we examined the age effects on the associated projection maps using the same correlation analysis. Notably, this analysis found an overall posterior-to-anterior-hippocampal connectivity shift during development, i.e., as children mature, the connectivity to the anterior part of the hippocampus increases, as compared to that of the posterior hippocampus (Fig. 4*B*). To evaluate the network-level preferential connectivity change, we used the Cole-Anticevic Brain-wide Network Partition (CAB-NP) whole-brain atlas (31) and found such preferential connectivity shift in a majority of networks such as DMN, frontoparietal, dorsal attention, cingulo-opercular, somatomotor, and visual networks, while nonuniform age-related changes were found in orbital-affective and auditory networks.

Multivariate Predictive Modeling of Hippocampal Projection Maps. Given the prior literature suggesting the importance of whole-brain integration of hippocampal longitudinal gradient for episodic memory (7), we decided to focus on the hippocampal projection maps. To study that, we built a phenotypic predictive model to investigate whether the projection maps of the primary gradients were neurobiologically useful. Considering the intricate preferential coupling pattern between the hippocampal gradient and the rest of the brain, univariate models may not be sufficiently sensitive to accurately depict the subtle and distributed brain-behavior relationship (32). Therefore, we utilized the least absolute shrinkage and selection operator (LASSO) model (33) to examine whether the primary projection maps can predict individuals' episodic memory, which ensures the generalizability of the model by imposing a penalty on the model complexity. Briefly, we first extracted the mean values of the primary projection maps from the HCP-D dataset using the CAB-NP whole-brain atlas (31),

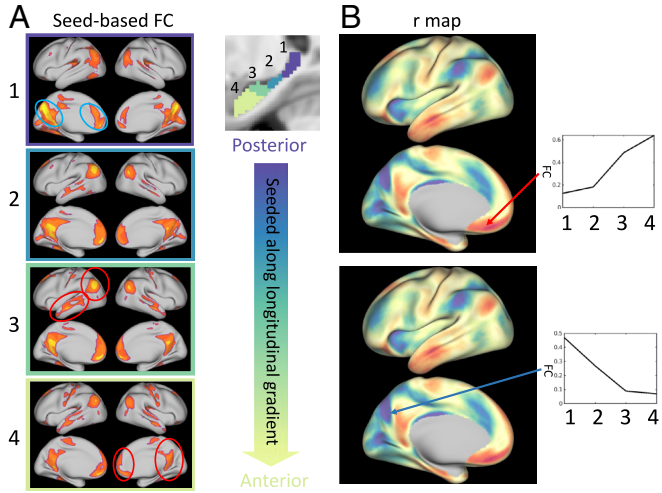

Fig. 3. (*A*) Seed-based functional connectivity along the hippocampal longitudinal gradient (seed regions are color-coded based on k-means clusters). For visualization purposes, only regions with the top 20% connectivity strength were shown. (*B*) Seed-based functional connectivity map as a function of seed location. Red reflects regions with increasing connectivity and blue for decreasing connectivity as the seed moves from the posterior to the anterior hippocampus.

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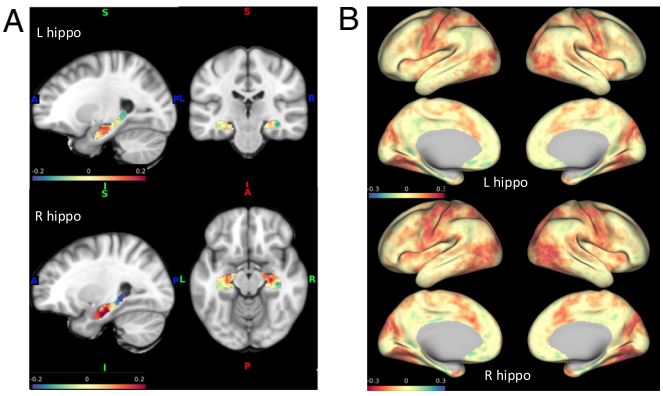

Fig. 4. Developmental analysis of the primary hippocampal longitudinal gradients with the HCP-D dataset. (*A*) Hippocampal age-related correlation maps for the left and right hippocampal gradient. (*B*) Age-related correlation maps of corresponding projection maps.

as shown in Fig. 5*A*. Using the LASSO model, we predicted the memory abilities measured by Picture Sequence Memory Test (34), with left hippocampal projection maps (r = 0.20, $p_{perm} < 0.001$, Fig. 5*B*). The predictive model remained significant after correction for head motion and age, suggesting that our memory predictive model was not driven by those factors. The predictive model using right hippocampal projection maps failed to reach significance after controlling for head motion. We then validated our memory predictive model in an independent dataset, successfully predicting the memory scores (r = 0.16, $P = 0.02$) measured by the Rey Auditory Verbal Learning Test in the NKI dataset (35).

For the memory predictive model trained using the HCP-D dataset, we observed a diffuse memory predictive pattern across the whole-brain. We observed strengthened connectivity between the left anterior hippocampus with the left temporal pole, left lateral orbitofrontal cortex, and left angular gyrus as well as between left posterior hippocampus and the left entorhinal cortex, right lateral orbitofrontal cortex, and right parahippocampal gyrus. Moreover, these regions mainly belong to DMN (n = 8), cingulo-opercular (n = 4), visual (n = 3), and orbito-affective networks (n = 3). Increased connectivity between the left *posterior* hippocampus and DMN, visual, and orbito-affective network is associated with higher predicted episodic memory scores. Increased connectivity between the left *anterior* hippocampus and the cingulo-opercular network is associated with higher memory scores. To test the specificity of our finding, we used the LASSO model with projection maps from the second hippocampal gradient and whole-brain hippocampal connectivity patterns, none yielding a significant prediction using the left or right hippocampus (Ps > 0.05).

To ensure that our memory prediction was not simply due to developmental effects, in addition to controlling for age effect in the previous analysis, we also built a predictive model using primary projection maps to predict individuals' age. We successfully predicted individuals' ages and achieved a significant prediction of r = 0.62 and 0.64 (Ps < 0.001) with the left and right hippocampal projection maps, respectively. We compared the model weights of the memory and age predictive models and found that the two model weights were dissimilar ($P > 0.05$), showcasing that the memory prediction was not driven by developmental effects.

Discussion

In this large-scale systematic examination of functional specialization along the hippocampal longitudinal axis in typically developing children, we adopted a well-established connectopic mapping technique to quantify the hippocampal gradients in a data-driven fashion. We confirmed that hippocampal functional organization in children followed a highly similar longitudinal pattern as previously found in healthy young adults (13) and elder adults (36). The hippocampal gradient also matched the hippocampal subdivisions of the Melbourne subcortical atlas (26). Our results provided additional evidence of the functional specialization along the hippocampal long axis, concordant with evidence from other modalities, including task fMRI (14), diffusion weighted imaging (15), gene expression (16), and multimodal imaging (17). Moreover, we showed age-related functional specialization along the long axis, characterized by increasingly distinct anterior and posterior hippocampal connectivity profiles. Additionally, we found that the general whole-brain connectivity preferentially shifts from the posterior to the anterior hippocampus as children age. Using a data-driven approach, we further delineated how the hippocampus is differentially integrated into the whole-brain cortical hierarchy in order to support episodic memory. Using a phenotypic predictive modeling framework, we demonstrated the key role of hippocampus within the dual PM/AT pathway. Using two large datasets within a rigorous predictive framework, we ensured the robustness and reproducibility of our findings.

When focusing on investigating the whole-brain projection patterns of hippocampal gradients, we found the projection patterns were also highly consistent with findings from previous literature using the connectopic mapping technique (13, 14). To unpack hippocampal projection patterns, we employed a moving-seed-based functional connectivity analysis by moving the seed from the posterior to the anterior hippocampus (r map). We found the r map—the level of changes of seed-based connectivity—highly mimics the hippocampal projection maps. Therefore, a region with a positive projection value can be understood as increasing connectivity while the seed moves anteriorly along the long axis and vice versa. Coinciding with previous functional connectivity studies (for a meta-analysis, see ref. 37), we found the anterior hippocampus to be more connected to the ventromedial prefrontal cortex (38), amygdala (39), entorhinal cortex (40), and anterior and lateral temporal lobe extending to the temporal pole (41). Meanwhile, the posterior hippocampus is preferentially connected to the inferior parietal lobule (42), anterior cingulate cortex (43), retrosplenial cortex (4, 43), precuneus (38, 43), and occipital gyrus (38, 43). Our projection pattern of the hippocampal long axis is further reinforced by the cortico-hippocampal structural connectivity patterns. For example, Dalton and colleagues showed that the occipital and medial parietal cortex (including precuneus and retrosplenial cortex) are highly structurally coupled to the posterior hippocampus, while the anterior hippocampus is more strongly linked to the temporal pole and lateral temporal cortex (15).

Looking more globally, we found evidence supporting two models that position the hippocampus into the whole-brain cortical network structure, the parallel hippocampal-parietal circuits (5) and the PM/AT framework (7). Starting with the posterior hippocampus, Zheng et al. found the posterior hippocampus connected to the PM parietal cortex (referred to as the parietal memory network by the authors), supporting goal-oriented cognition and stimulus recognition (5). The PM parietal cortex mainly encompasses the strip of the posterior midline neocortex between motor and visual regions and partially overlaps with the posterior DMN. Also, part of a putative posterior memory network (44), this region has demonstrated a selective vulnerability in epilepsy patients showing altered hippocampal functional connectivity (45). The differential coupling pattern between the hippocampus from the DMN is also evident in other studies. For example, Barnett and colleagues showed that DMN consists of

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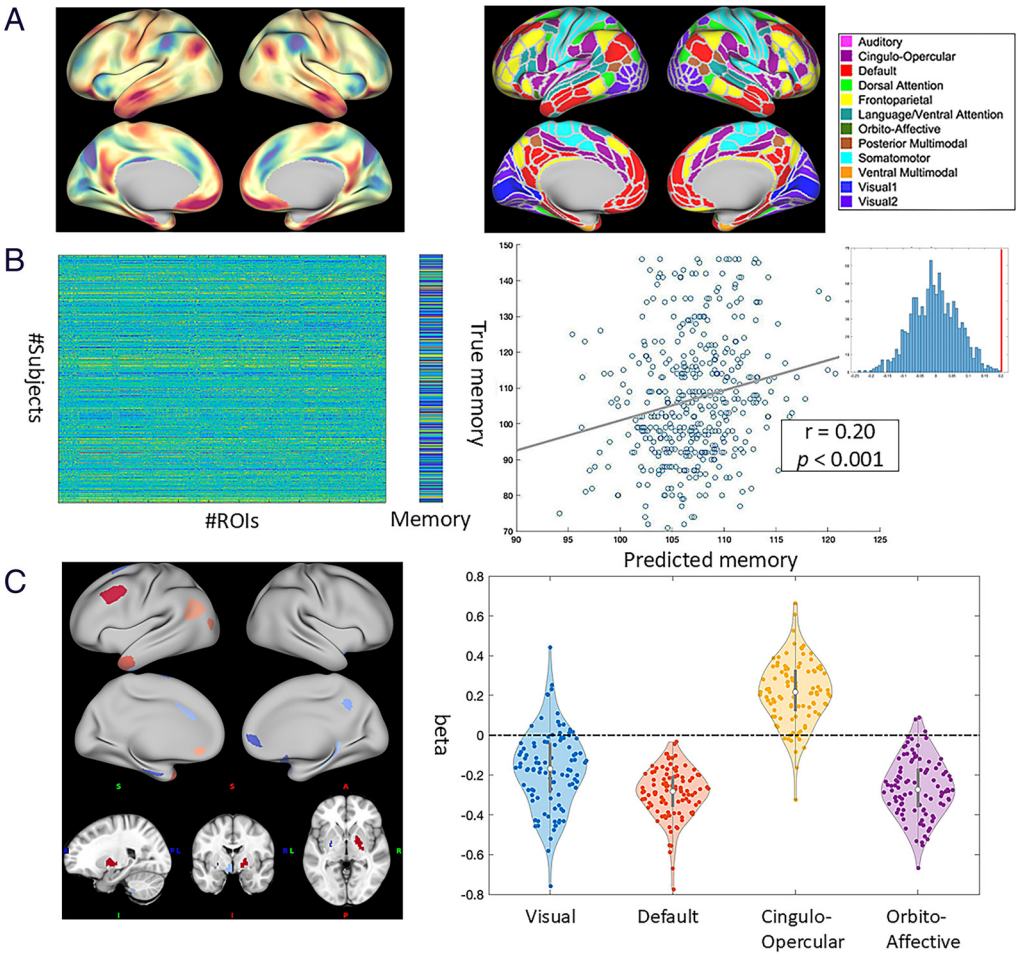

Fig. 5. A graphical overview of the predictive framework. (*A*) Projection values from the primary projection maps were extracted using the CAB-NP whole-brain atlas. (*B*) The extracted mean projection matrix was entered into a LASSO model, significantly predicting the memory scores in the left-out validation set (r = 0.20, $p_{perm} < 0.001$). (*C*) The memory-predictive pattern for left hippocampal projection maps and the violin plot of model weights for each network across 10 × 10 CVs. Positive weights mean increased connectivity to the anterior hippocampus, while negative weights mean increased connectivity to the posterior hippocampus.

multiple subnetworks (4), including a PM subnetwork (mostly within the PM parietal cortex). The PM subnetwork, together with the posterior hippocampus and parahippocampal gyrus, forms a so-called PM cortical pathway (7). The PM system is responsible for processing context information, scene perception, and episodic and autobiographical memory (6). By contrast, the anterior hippocampus preferentially coupled with the entorhinal cortex, perirhinal cortex, amygdala, and lateral orbitofrontal cortex, forming the AT cortical pathway. As part of the theorized PM/AT framework, the AT pathway supports recognition and associative memory, affective, and semantic processing (6). We also observed stronger coupling between the anterior hippocampus and the remainder of the DMN (e.g., angular gyrus, medial temporal lobe, medial prefrontal cortex, and posterior cingulate cortex), which could be part of the neural circuitry supporting social cognition, emotions, and self-referential judgments (5). Our findings complemented the two models by situating the longitudinal hippocampal gradient in large-scale cortical systems and further reinforced the role of hippocampus as an integration hub facilitating communication between brain networks for the convergence and joint processing of information (46).

Besides largely replicating findings from adult studies regarding the longitudinal hippocampal functional specialization and its whole-brain preferential coupling pattern, we also further explored their developmental effects. The age-related gradient correlation maps, which highly resembled the primary hippocampal gradients, demonstrated that the patterns of anterior and posterior hippocampal connectivity become increasingly distinct throughout development, suggesting that stronger functional specialization emerges along the hippocampal long axis as brains mature. Our finding of the overall posterior-to-anterior preferential connectivity shift contrasts with an earlier seed-based connectivity study, in which Blankenship et al. found overlapping connectivity patterns for anterior and posterior hippocampus (47). This discrepancy may result from their choice of hippocampal seeds, as both anterior and posterior seeds included the parts of hippocampal body, a region known for having a relatively poor functional segregation (25). Including this area may therefore diminish functional specificity, whereas the gradient method used in our study rules out this technical issue. Furthermore, the observed posterior-to-anterior preferential connectivity shift aligns with previous structural and functional neuroimaging studies that suggest a potential developmental lag of the anterior hippocampus as compared to the posterior hippocampus (20, 48–50). This can be interpreted through the potential functional specialization along the hippocampal long axis, given the anterior hippocampus's critical role in the flexible retrieval of gist-like information and the posterior hippocampus's role in reconstructing detailed, precise information (25). It is possible while younger children primarily rely on reinstating exact previously encoded representations during episodic memory retrieval, adults and older children may more flexibly merge contextual information into the gist-like concepts, a process heavily

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involving the anterior hippocampus (48). Our finding of the increased integration of the anterior hippocampal region during the later period also points to its involvement in more mature and sophisticated cognitive functions (e.g., schema).

In addition, we also built multivariate models to explore the relationship between hippocampal projection maps and individuals' episodic memory abilities. We successfully predicted out-of-sample memory ability using LASSO in the discovery dataset and further validated our models in an independent test set. Efficient communication among large-scale brain networks, with the hippocampus serving as a hub, is critical for memory (51). To further delineate the role of functional specialization along the hippocampal long axis in memory, researchers have conducted extensive research on the differential hippocampal–cortical coupling pattern for anterior vs. posterior hippocampus (36, 39, 42, 52–54). Our memory predictive model extended these earlier findings, as we observed a diffuse predictive pattern featuring the hippocampal gradient-related neural coupling shift with a few key canonical brain networks, such as cingulo-opercular and visual networks, with the DMN contributing the most (Fig. 5*C*). Situated at the top of the whole-brain functional hierarchy in between unimodal to multimodal cortices (12), the DMN serves as a foundation that can optimally facilitate the integration of rich and complex semantic and episodic information processing (55). The central role of DMN may explain the high prevalence of DMN nodes within our memory predictive model. Interestingly, we observed an overall shift of DMN toward the posterior hippocampus. The integration of the DMN and the posterior hippocampus was noted in adults, which is supposed to be tied to memory retrieval (56). Taking a closer look at the predictive regionof interests (ROIs), we found many were also key nodes within the PM/AT framework. For instance, we observed better episodic memory prediction was associated with strengthened coupling between the left anterior hippocampus and left temporal pole, left lateral orbitofrontal cortex, and left angular gyrus. Moreover, higher predicted memory was related to stronger coupling between the left posterior hippocampus and left entorhinal cortex, right lateral orbitofrontal cortex, right parahippocampal gyrus, and right posterior cingulate cortex. The predictive model relationship with memory was not found with the right hippocampus may be due to the left hippocampus being dominant for language and episodic memory (57, 58). Despite the use of pictures used to probe episodic memory in the HCP-D dataset, it is also plausible that participants still rely on verbal strategies to encode and retrieve information, given that it is a sequencing of pictures that are verbally described images of everyday situations (e.g., how to go camping). Thus, leading to significant findings for the left hippocampus with memory performance, given its role in verbal memory (59).

Limitations and Future Directions. A potential caveat of our study is that the limited spatial resolution and smoothing kernel used in the preprocessing prevented us from delineating the contribution of hippocampal anatomical structure to the longitudinal functional gradients, e.g., hippocampal subfield divisions (60). High-resolution fMRI at high field strength together with surface-based hippocampal analysis [e.g., unfolding the hippocampus to a two-dimensional sheet (61)] will help better (de)couple its structural and functional organization. Here, we only focused on the primary longitudinal gradient, while studies have suggested that more than one gradient could be of interest (62, 63). Although in the current study, we did not find any association between the projection maps of the second hippocampal gradient in terms of memory prediction, potentially because the second gradient may be linked to hippocampal local microstructural properties (60). Future studies should explore the functional significance of higher-order gradients. Furthermore, in light of the multifaceted nature of the hippocampus (64), it would be interesting to test whether our current framework can be generalized to domains beyond memory. Given the critical role of the hippocampus in neurological diseases, it would be interesting to investigate the hippocampal gradients and their associated whole-brain coupling pattern in clinical populations, such as dementia and epilepsy. Using large-scale clinical samples, future studies can examine the hippocampal gradient for its link with risk factors and treatment outcomes. Moreover, despite the numerous benefits of using large-scale open datasets, it comes with its unique challenges, such as diverse cognitive assessments. Although Picture Sequence Memory Test used in the HCP-D dataset shows high correlation with other standardized episodic memory tests such as Rey Auditory Verbal Learning Test used in the replication dataset (34), working memory processes may play a part in the task performance of Picture Sequence Memory Test, given the relatively short delay between encoding and retrieval (65). However, our significant prediction accuracy (further corroborated by its generalizability in an independent replication dataset) suggests that our model at least properly captured the hippocampal basis serving basic sequence memory functions. Future work may be expandable to more robust and homogenized assessments. Finally, although we have shown monotonic age-related changes in hippocampal gradients and projection maps using cross-sectional samples, only longitudinal studies can definitively elucidate the maturation trajectory of longitudinal gradients and their precise relationship with the development of cognitive functions (66).

Conclusions

Leveraging large-scale open neuroimaging datasets, we identified a primary longitudinal hippocampal gradient, which we situated within the large-scale cortical system in developing brains. By building a reproducible memory predictive model using hippocampal projection maps, we combined gradient methodology with phenotypic predictive modeling and demonstrated how differential integration of the longitudinal hippocampal gradient supports individuals' episodic memory. Our results showed the posterior hippocampus was embedded in a PM pathway, while the anterior hippocampus was integrated with the AT pathway, extending the PM/AT framework to children (7). Throughout development, we observed a strengthening of functional specialization along the hippocampal long axis, accompanied by a whole-brain preferential connectivity shift from the posterior to the anterior hippocampus. Our memory predictive model identified several key nodes within the PM and AT pathways, such as the entorhinal cortex and parahippocampal gyrus, demonstrating the key role that the PM/AT dual pathways play in episodic memory. In summary, our results emphasize the key role for the longitudinal gradient and its cortical interaction in both development and the support of episodic memory.

Materials and Methods

Participants. Two publicly available resting-state fMRI datasets were used in the study. The primary discovery dataset, HCP-D (22), consists of 652 typically developing children and adolescents between the age of 5 to 21. Among them, 598 participants between 8 and 21 remained after excluding those with high motion (mean framewise displacement > 0.4 mm), incomplete scans (less than four runs), focal anomalies (QC_Issue_Codes = A), or unsuccessful surface reconstruction. The enhanced NKI-Rockland Sample (21) is an open large-scale community-ascertained lifespan dataset, which is used as a replication dataset to

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evaluate the reproducibility of our predictive models. The same exclusion criteria were applied, and a total of 196 participants remained with memory scores.

Image Acquisition and Preprocessing. For the HCP-D dataset, the participants were scanned on a 3T Siemens Prisma with a 32-channel head coil. The T1-weighted multiecho Magnetization-Prepared Rapid Acquisition Gradient Echo images were collected with the following scanning parameters: 0.8 mm isotropic resolution; repetition time (TR) = 2,500 ms; echo time (TE) = 1.81:1.79:7.18 ms; flip angle = 8°. The functional data were acquired using a multiband echo planar imaging sequence with the following parameters: 2 mm isotropic resolution; TR = 800 ms; TE = 37 ms, flip angle = 52°, multiband acceleration factor = 8. A total of 26 min of resting-state scans was collected as participants were instructed to stay still and awake, and look at the fixation crosshair. For the NKI dataset, T1-weighted images were collected with the following scanning parameters: 1 mm isotropic resolution; TR = 1,900 ms; TE = 2.52 ms; flip angle = 9°, and the functional scans were collected with the following parameters: 3 mm isotropic voxels, TR = 645 ms, TE = 30 ms, flip angle = flip angle = 60°, multiband acceleration factor = 4. The participants were asked to keep their eyes fixated on a crosshair, stay still, and not think of anything.

A standard preprocessing toolbox (fMRIPrep v 22.0.2) (67) was used to preprocess the imaging data. Briefly, the functional data were registered to T1w structural data after susceptibility distortion correction, slice-timing correction, and motion correction. We applied a high-pass filter (>0.01 Hz) and removed the linear and quadratic trends, and nuisance regressors (six motion parameters, mean signal of ventricular and white-matter signal, and their temporal derivatives). The denoised data were concatenated across runs and then projected to a combined volume and surface-based analysis format ("MNINonLinear_fs_LR_32k") using ciftify tools (68), followed by spatial smoothing across each surface independently using a 6 mm full width at half maximum Gaussian kernel.

Hippocampal Connectopic Mapping and Projection Map. We employed the connectopic mapping analysis (https://github.com/koenhaak/congrads) (11) to characterize the functional gradient describing the topological changes in the connectivity organization of a ROI. The connectopic mapping and similar methods have been used to accurately capture the functional architecture of various brain regions, such as retinotopic and somatotopic cortex (11), hippocampus (13, 60), thalamus (69), and subcortex (26). Here, we investigated the hippocampal gradient and its functional coupling pattern to the entire cortex. Briefly, we started by applying principal component analysis on the denoised fMRI data. We then computed whole-brain functional connectivity between all voxels in the hippocampus (as defined by Connectivity Informatics Technology Initiative built-in hippocampus label) and the principal component analysis-transformed data using Pearson correlation. From the connectivity matrix, we calculated a similarity matrix among hippocampal voxels using η^2 coefficients for left and right hippocampus separately. The η^2 coefficients measure the similarity of voxelwise functional connectivity, ranging between 0 (entirely dissimilar) to 1 (entirely similar). The similarity matrix was transformed into a sparse adjacency matrix. Laplacian Eigenmap (24), a nonlinear dimensionality reduction method preserving the local geometry of the data, was applied on the sparse adjacency matrix, so that voxels with similar connectivity profile were close to one another in the low-dimensional space. The resulted gradient provides a mapping of continuous variation in whole-brain functional connectivity across the topography within the hippocampus. Following the estimation of the group-level gradients from the averaged adjacency matrices of all subjects, the individual-level gradient was aligned to the group-level gradients using Procrustes analysis. Given the theoretical and empirical significance of functional segregation along the hippocampal longitudinal axis (10, 13, 25), we focused on the primary longitudinal gradient and its corresponding projection map. Each subject's hippocampal projection map was calculated from the aligned individual-level gradient using dual regression (28), which reflects the preferential functional coupling patterns between the topologically varying gradient and the rest of the brain (27).

Projection Maps Predicting Memory Abilities. To investigate the biological meaning of projection maps and better understand how hippocampus whole-brain interaction supports memory, we examined whether hippocampal projection maps could be used to predict episodic memory abilities. For memory abilities, the total number of correctly positioned adjacent pairs of pictures for the NIH Toolbox Picture Sequence Memory Test (34) was used for HCP-D participants, and the delayed recall score of Rey Auditory Verbal Learning Test (35) was used for NKI participants. The former was age normalized and the latter was age and gender normalized. For the predictive model, we used a sparse LASSO model. We first parcellated the individuals' projection maps using the CAB-NP whole-brain atlas (31), which extends the cortical networks to the subcortex, cerebrum, and brainstem. After excluding parcels with less than ten voxels/vertices, the mean projection values of the remaining 543 ROIs were entered into a 10 × 10-fold nested cross-validation (CV). Briefly, the discovery HCP-D dataset was initially partitioned into two sets, i.e., 90% training set and 10% test set. The inner loop used the training set to determine the optimal model and hyperparameter. In the outer loop, the optimal model was assessed on the test set to provide an unbiased evaluation of a final model fit, and the process was repeated ten times. We reported the average performance of the ten outer loops. The optimal models from the HCP-D dataset were averaged with the most consistent features chosen (features included in more than 95% CV folds), and the final averaged model was assessed in the replication NKI dataset.

Null Analysis of Hippocampal Projection Maps. We also utilized a null model to test whether projection maps could arise due to various nuisance factors, such as spatial autocorrelation (30). Specifically, we replaced 100 subjects' hippocampal timeseries with those from random subjects to maintain the local spatial autocorrelation and local functional connectivity structures and repeated the connectopic mapping analysis on the synthesized random data. We then compared the across-subject similarity of projection maps derived from the real and random data. Additionally, we wanted to understand how much cortical geometry influenced the functional gradients and corresponding projection maps. We characterized the anatomical gradients by performing principal component analysis on the hippocampal voxel coordinates in the MNI space and compared the anatomical gradients with functional gradients.

Data, Materials, and Software Availability. The data supporting the results in this study are available within this paper and its *SI Appendix*. The HCP-D dataset is publicly available at the National Institute of Mental Health Data Archive (70), and the NKI dataset is publicly available through International Neuroimaging Data-sharing Initiative from the Nathan Kline Institute https://fcon_1000.projects.nitrc.org/indi/pro/nki.html (71). Previously published data were used for this work (21, 72). The software used for the gradient analysis is publicly available from GitHub repository congrads (73).

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